

GENESIS

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Figure 1: Carbon – Prehistoric jungle with giant insects

ABSTRACT

GENESIS embarks on an emotionally intense virtual reality journey to experience the dramatic milestones in the evolution of Earth and mankind. As a 360°, stereo 3D film, GENESIS has been pre rendered (no real time) and is usable as a three Degrees of Freedom (3DOF) experience. The project can be watched on all VR glasses that can play 3DOF, like 360° movies. Glasses such as Pico G2 or Oculus Go were developed specifically for 3DOF content. Of course, all glasses which are able to present 6DOF experiences can also play 3DOF content. The users can watch the content while standing or sitting.

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1 CONTENT

We are all familiar with analogue film images about the history of our earth. However, through immersion you experience history in a completely new way - you live these rebirths, these outbreaks of life. All those drastic events in Earth's evolution.

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Chaos, rebirth and catastrophes mark the history of Earth, yet mankind exists. It was our goal to highlight the most dramatic milestones in the evolution of Earth and mankind in our film GENESIS. We let people immerse themselves as time travelers into unknown spheres. To discover unreal, mystical and daunting worlds, full of breath-taking landscapes and creatures that truly existed at those times.

At the beginning, the viewers see the young, glowing Earth surrounded by cosmic debris. Then viewers see the Earth flying over endless oceans of darkness before moving through a magical underwater world. Emerging from the water an enchanted land unveils: a paradise of prehistoric jungle and giant insects, followed by mass extinction – until a new cycle starts that surrounds us with colossal dinosaurs who conquer the Earth, before meeting extinction themselves. Then finally, viewers come face-to-face with our human ancestors, catching glimpses of their achievements and our legacy. What remains is an intense impression of the Earth and the insurmountable force and fragility of life.

When placing all of the selected highlights within a timeline, we became aware that we needed an analogy - an analogy that emphasizes that the majority of milestones happened in the most recent phases of Earth's history.

There's one time relation we are most familiar with: the 24-hour analogy. We were drawn to create an intense new perspective on the unimaginable dimension of time and imagined a chronometer which actually needed to be changed to different time scales twice. Once from hours to minutes, and then from minutes to seconds

Designed in great detail, we refined the graphics and also the movement of the time wheel several times until everything became as obvious for the viewer as possible. The combination of the scenes



Figure 2: Virtual Whiteboard for GENESIS

and the chronometer happened to be essential. At this moment, everything came together.

4.7 bn years of evolution condensed into a 24 h metaphor reveal: We are children of galactic luck, born in the final second of eternity.

2 IMPLEMENTATION

Due to the history of the Faber Courtial studio, the production of 3D animation and VFX for the film and television industry was fixed from the beginning on the typical VFX pipeline. In our case, that meant 3D creation via the 3DS Max 3D software, rendering of each layer, afterwards compositing and the further post-production. A complicated and laborious work process.

During the early production of GENESIS, it was decided to switch the pipeline and continue producing GENESIS in the real-time engine Unreal. A difficult decision at the time, but the quality in Unreal had improved massively in 2020. The quality was no longer inferior to production with a traditional pipeline. And by producing in a real time engine, all possibilities are still open for later further formats like 6DOF.

We stayed with our decision of producing GENESIS initially as a VR film – we are familiar with this workflow and had to work within a tight schedule. Even so we made a lot of tests before our decision to change the pipeline, we had several unpleasant surprises during the production.

Unreal 4 (back then a beta version) so far wasn't adapted for the export of 360 film. A film consists of tens of thousands 360° stereo 3D images which now had to be rendered in Unreal. Our hope was, that it should be much faster in a real time engine, which wasn't the case. And the image calculation resulted in many errors:

With a traditional pipeline, the calculation of an image is based on the algorithm of raytracing: The calculation happens in a pixel-accurate manner, a very clean method. In a game engine, a 360 image is created like it would be by a 360 camera with different camera lenses in different directions: so, in the end you have many images which have to be stitched.

This leads to many stitching problems within the production of GENESIS and the quality of the created images wasn't convincing. We tested different tools and compared their features: the required time for rendering and the quality of the images. We also wanted to use post effects like lens flare and Bloom, which couldn't be used in some of the available rendering tools. In the end we opted for the plugin ANSEL. The quality of the images seemed convincing, so we expanded it by programming to use it also for a complete sequence not just one image (original purpose).

Moving objects within rendered sequences presented another problem. They contained a kind of jitter. Finally, we found the cause in the motion vector within the real-time engine.

Apart from creating high quality and intense sequences, we put a lot of emphasis on a very fine interweaving of images, sounds, music and voice over during post-production.

3 REACTIONS

Genesis had its international premiere at the Venice Film Festival 2021 and the North American premiere at SXSW. It was selected and had won the Crystal Owl Award at Stereopsia 2021 for Best 360 Video. The film continues to be requested for many festivals and is very popular.

Throughout Genesis is received as a remarkable way of following along the development of life on earth. Viewers and jurors describe it as „lively and charismatic“, „enticing“ and „high quality in every detail“.